CompSci 471

Artificial Intelligence

Lesson 7: Heuristics

Estimate → Verify → Compare → Guarantee

$h(n)$ → $h^*(n)$

estimate | true cost

$$h(n) \le h^*(n)$$

Quality test:
Is it... safe, coherent, informative, and worth computing

Better guidance can make search faster, but unsafe guidance can remove the guarantee.

Fall 2026

Quiz 1

10 minutes. Individual work. Stop when called.

Before you begin
Put away unauthorized materials
Write name and section
Read each instruction carefully

During the quiz
No collaboration
No devices
Show work only where requested

After time
Stop immediately
Pass papers forward
Transition to heuristic quality

Quiz 1 serves as retrieval practice for Lessons 4–6.
Today's new content begins after collection.

Reset: from using heuristics to judging them

Last lesson asked how A* uses h(n). Today asks whether A* should trust h(n).

$$f(n) = g(n) + h(n)$$

g(n)
Exact cost already incurred from the start to n

h(n)
Estimated remaining cost from n to a goal

Quality of h(n)
Controls efficiency and reliability

Bridge question: what could go wrong if the estimate is too aggressive?

Today’s mission

By the end of Lesson 7, students should be able to…

1. **Define admissibility, consistency, and dominance**
2. **Evaluate heuristic quality and its impact on performance**
3. **Explain why A* is optimal under certain conditions**

Performance target: audit a heuristic table, check one edge, compare two heuristics, and justify A*’s guarantee.

The missing reference: $h^*(n)$

To judge an estimate, compare it with the true optimal remaining cost.

$h^*(n)$ = true least cost from n to a goal

$h(n)$

The heuristic estimate used by the algorithm.

$h^*(n)$

The actual optimal remaining cost.

In real problems, $h^*(n)$ is usually unknown.
In classroom examples, it lets us evaluate the heuristic.

A spectrum of heuristic behavior

More guidance is helpful only while the estimate remains safe.

0 estimate
safe
least guidance

weak lower bound
safe

strong lower bound
safe
more guidance

exact
safe
perfect

overestimate
unsafe
guarantee lost

A zero heuristic is safe but unhelpful. A perfect heuristic is safe but may be hard to compute.

Admissibility: never overestimate

An admissible heuristic is optimistic about the remaining cost.

$$h(n) \leq h^*(n) \text{ for every state } n$$

Allowed
Underestimate
Equal the true cost
Return zero

Not allowed
Overestimate at even one state

Interpretation
The heuristic may be conservative, but it cannot claim the goal is more expensive than it truly is.

One overestimate is enough to make the heuristic inadmissible.

Admissibility audit

Check every state, not the average behavior.

State	$h^*(n)$	$h_1(n)$	$h_2(n)$
P	9	5	8
Q	6	4	7
R	4	2	4
G	0	0	0

Questions

1. Is h_1 admissible?
2. Is h_2 admissible?
3. Which state determines the answer?

Reveal

h1 is admissible.

h2 is not: Q overestimates because 7 > 6.

One successful search proves very little

A good outcome on one run does not prove the heuristic is safe.

"A* returned the optimal route, so the heuristic must be admissible."

Why false?

An inadmissible heuristic may still return the optimal solution on a particular graph or run.

What we need

Admissibility is a property of the heuristic over all relevant states.

Evidence from one run can be encouraging, but it is not a guarantee.

Consistency: estimates must agree locally

A heuristic should behave sensibly across every edge.

$$h(n) \leq c(n,n') + h(n')$$

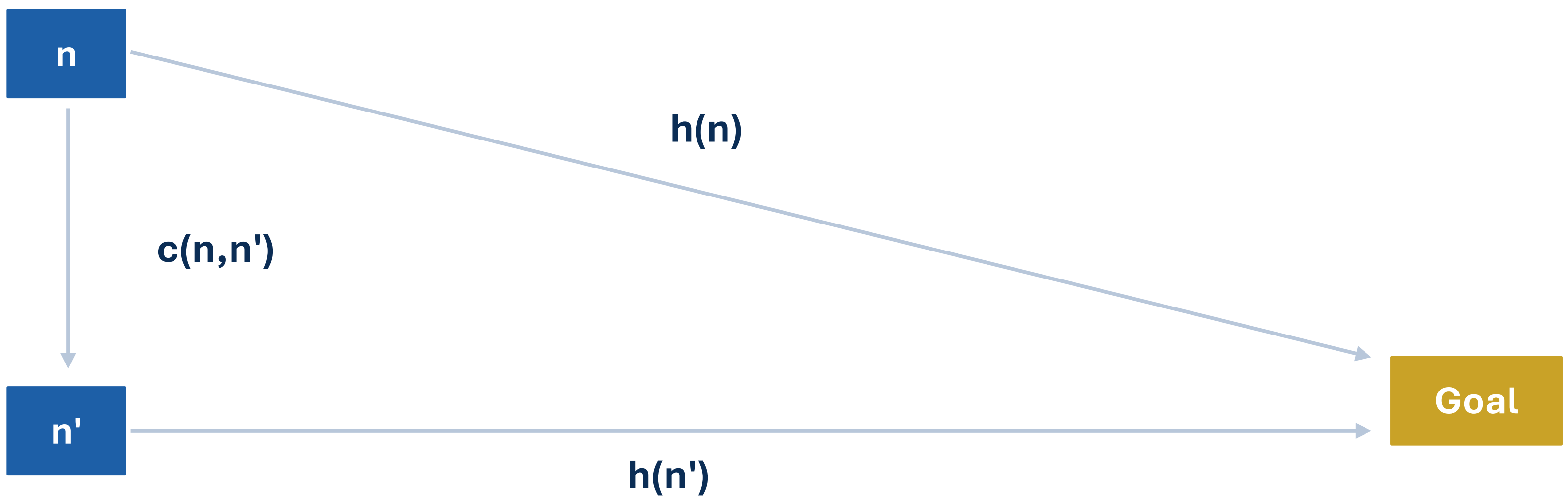
Meaning
The estimate at n cannot exceed one step to n' plus the estimate from n'.

Memory hook
The heuristic cannot drop by more than the edge cost.

Consistency is checked edge by edge.

Consistency triangle

Compare the direct estimate with one step plus the next estimate.



direct estimate from n ≤ one step + estimate from n'

Edge-by-edge consistency check

Substitute values into the inequality.

Given edge X → Y
h(X)=8
c(X,Y)=3
h(Y)=4

$$8 \le 3 + 4$$

$$8 \le 7 \quad \textbf{False}$$

Conclusion
The edge is inconsistent.
The estimate at X is too high relative to successor Y.

If consistency is violated, the estimate at the earlier state is too high.

Admissibility and consistency are related

They are both safety checks, but they compare different things.

Admissibility	Consistency
Compares h(n) with true remaining cost $h^*(n)$	Compares neighboring heuristic values
Global safety condition	Local edge condition
Never overestimates the goal distance	Cannot fall too sharply across an edge

Relationship

With h(goal)=0, every consistent heuristic is admissible.
But an admissible heuristic is not automatically consistent.

Why consistency matters to A*

With consistency, f-values do not decrease along a path.

$$g(n') = g(n) + c(n,n')$$

$$h(n) \leq c(n,n') + h(n')$$

$$f(n') = g(n') + h(n') \geq g(n) + h(n) = f(n)$$

Interpretation

Once A* has closed a state under a consistent heuristic, it does not later discover that the state should have had a lower priority.

State A* guarantees precisely

The version of A* matters.

A* version	Sufficient heuristic condition
Tree-search A*	Admissible heuristic
Standard graph-search A*	Consistent heuristic

Important caution

An overestimating heuristic may still work on a particular graph, but the optimality guarantee is lost.

Guarantees depend on assumptions. State the condition, not just the algorithm name.

Dominance: more information without losing safety

Among admissible heuristics, larger lower bounds can guide A* better.

If $h_2(n) \geq h_1(n)$ for every n, and both are admissible, then h_2 dominates h_1.

Safe
h_2 remains a lower bound and does not overestimate.

Informative
h_2 is at least as close to $h^*(n)$ everywhere.

Efficient
A* with h_2 generally expands fewer or equal nodes.

Dominance only applies after safety has already been checked.

Dominance comparison

Check safety first, then compare informativeness.

State	$h_1(n)$	$h_2(n)$	$h^*(n)$
P	2	6	7
Q	3	5	6
R	1	3	4
G	0	0	0

Questions

1. Are both admissible?
2. Does one dominate?
3. Which should usually guide A* more efficiently?

Reveal

Both are admissible. h_2 dominates h_1 because it is never smaller and is sometimes larger.

“Larger” is not automatically “better”

Better means more informative while preserving the needed guarantee.

Incomplete rule
Choose the heuristic with the larger values.

Correct rule
Among admissible heuristics, prefer larger lower bounds if the computation is worthwhile.

A heuristic can be...
larger but unsafe
safe but expensive
easy but weak
strong only for one representation

Three dimensions of heuristic quality

A good heuristic is safe, informative, and worth computing.

Safety
Is it admissible and, for graph search, consistent?

Information
How close does it get to $h^*(n)$ without exceeding it?

Cost
How expensive is the heuristic itself to compute?

Reducing node expansions is valuable only when the guidance does not cost more than the search effort it saves.

Where safe heuristics come from

One common strategy is to solve an easier relaxed problem.

Remove a constraint → solve easier problem → use that cost as a lower bound

Navigation
Ignore obstacles → straight-line distance

Terrain
Ignore restricted terrain → direct traversal estimate

Scheduling
Ignore conflicts → simplified completion time

Maneuvering
Ignore constraints → idealized energy estimate

The relaxed problem is easier, so its optimal cost cannot exceed the real optimal cost.

Heuristic audit

Pairs: recommend a heuristic and defend the recommendation.

Task 1
Identify any admissibility violation.

Task 2
Check two assigned edges for consistency.

Task 3
Determine whether one heuristic dominates another.

Task 4
Recommend a heuristic for A*.

Task 5
Defend using safety, information, and cost.

Do not only say which heuristic is bigger.
Explain why it is safe, useful, and worth computing.

Diagnose the claims

Correct each statement before it becomes a misconception.

1. **A heuristic is admissible when it is correct most of the time.**
2. **Consistency compares $h(n)$ directly with $h^*(n)$.**
3. **The heuristic with the largest values always dominates.**
4. **If an inadmissible heuristic works once, A* is still guaranteed optimal.**
5. **A zero heuristic is unsafe because it is not informative.**

Synthesis: heuristic quality

Rebuild the core tests from memory.

Admissible

$h(n)$ ______ $h^*(n)$

Consistent

$h(n)$ ______ $c(n,n')+h(n')$

Dominance

$h_2(n)$ ______ $h_1(n)$, with both admissible

? ≤ ?   ? ≤ ?   ? ≥ ?

Three questions

Is it safe?
Is it informative?
Is it worth computing?

Exit Ticket

Apply the heuristic-quality audit to one compact case.

State	$h^*(n)$	h(n)
A	8	6
B	5	7
C	3	2
G	0	0

Answer

1. Is h admissible?
2. Which state determines the answer?
3. Why does one successful A* run not prove safety?

Next class

Adversarial and stochastic decision-making: the environment begins responding as an opponent or through chance.